

CI 345: Machine Learning Research Paper

Azzam Mehdi, Abdul Wasey, Arsh Vahora

East-West University, Department of Computer Science and Information

Chicago, Illinois, 816 S Michigan Ave, 60605

me24624@eastwest.edu, mo23266@eastwest.edu, va23727@eastwest.edu

GitHub Link: <https://github.com/Wasey23/CI-345-Machine-Learning-Project>

Abstract

This research paper examines whether machine learning techniques can identify patterns associated with financial success in the movie industry. By analyzing historical movie data, classification models can learn relationships between movie characteristics and past outcomes, and use them to predict whether a movie is likely to succeed at the box office. To investigate this problem, six classification algorithms were created to predict box-office success.

Introduction

Movies have long been an extraordinarily popular medium for people around the world to enjoy entertainment. As audiences have continued to support films in theaters, on DVDs, via streaming services, and on other platforms, the film industry has grown into a multi-billion-dollar business. The revenue generated by successful films is reinvested in future productions, allowing directors, actors, producers, and others in the industry to create many more films. Unfortunately, due to factors such as audience preferences, marketing effectiveness, release timing, and competition from other films, it is often uncertain whether a movie will succeed at the box office. Many studios invest millions of dollars across production, marketing, and distribution, yet there is no guarantee that a movie will generate enough revenue to be considered successful. Because of this uncertainty and the associated financial risks faced by many in the industry, forecasting box-office performance has become an important problem for both industry professionals and researchers.

This paper examines whether machine learning techniques can identify patterns in a movie's success and answers the question: Can different classification models predict box-office success by analyzing historical movie data? By uncovering relationships between specific movie characteristics and past outcomes, these models can predict whether a movie is likely to achieve box office success.

Dataset Description

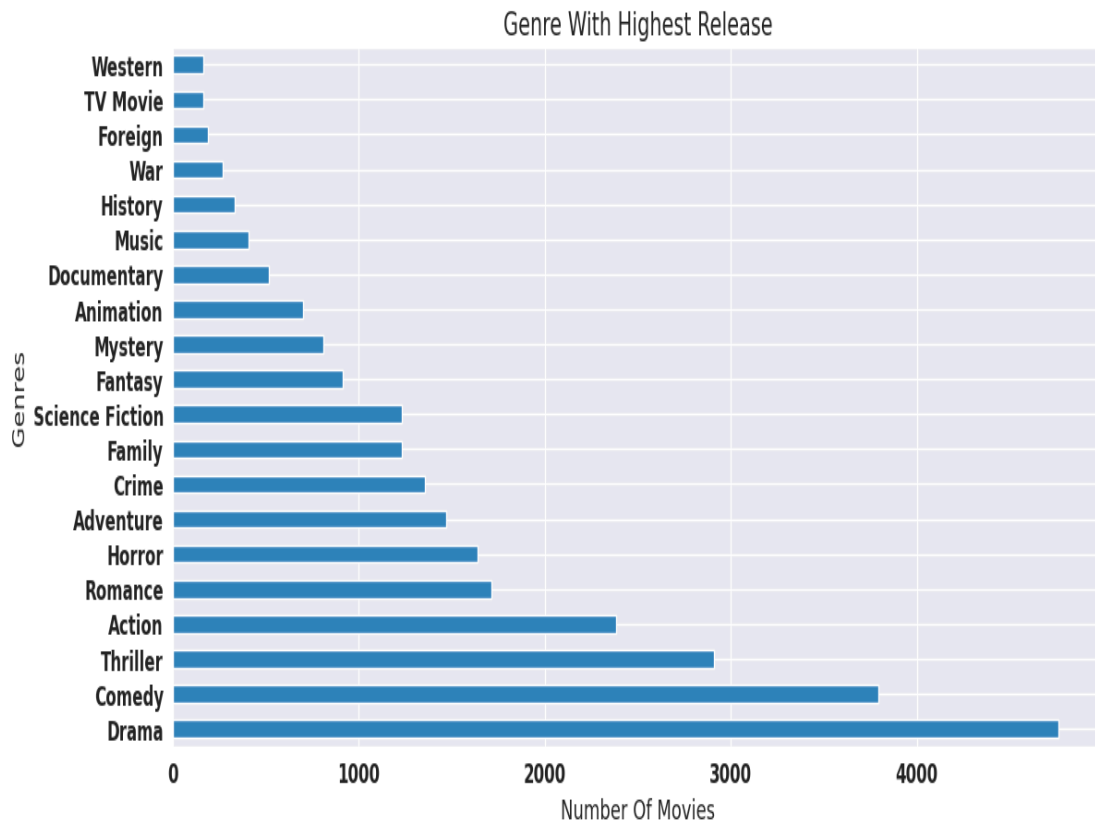
This study utilizes the TMDb Movie Dataset, which contains information on approximately 10,000 movies released between 1960 and 2015. Our dataset was selected from The Movie Database (TMDb), a popular online database that provides information regarding films, television programs, and related media. The dataset includes several attributes for each movie, such as budget, revenue, popularity, runtime, and release date.

This dataset was selected for its large and diverse collection of movies. In addition, including features such as budget, revenue, and popularity provides information for examining factors that may influence a movie's performance. These traits make the dataset well-suited for machine-learning classification and prediction of box-office outcomes.

To transform the problem into a binary classification task, a new target variable, *is_hit*, was created. Movies were labeled as box-office hits (*is_hit* = 1) if their revenue exceeded twice their production budget, whereas movies whose revenue was less than or equal to twice their production budget were labeled as flops (*is_hit* = 0). This threshold was selected because generating revenue that more than doubles the production budget is a commonly used industry benchmark for commercial success, as it accounts for both direct production costs and substantial marketing and distribution expenses not captured in the production budget alone. By applying this criterion, the target variable offers a defensible and practical distinction between commercially successful and unsuccessful films.

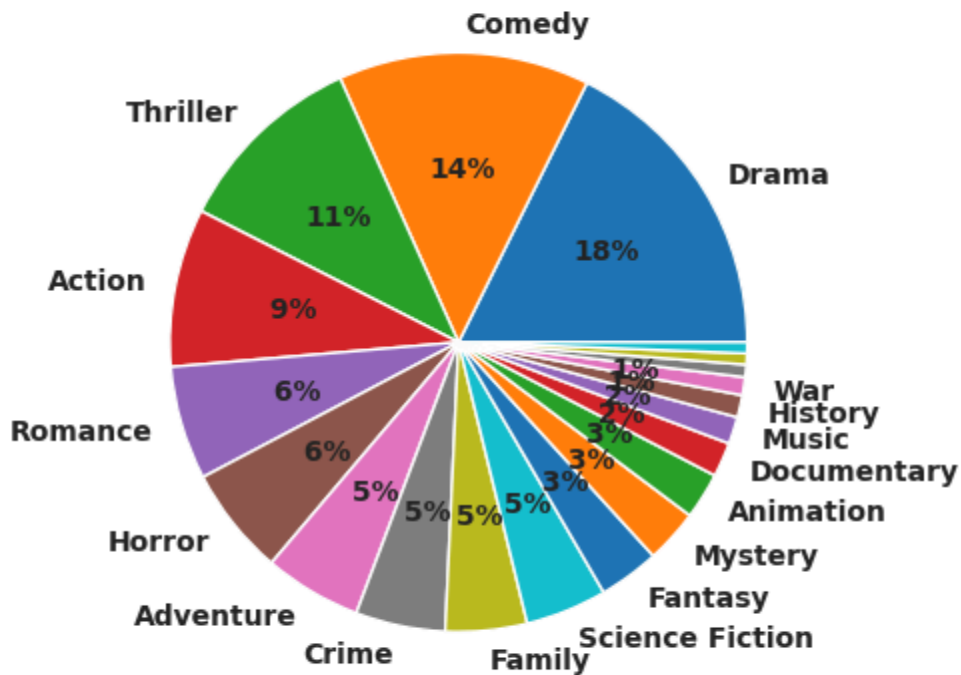
Data Analysis and Visualization

The TMDb dataset used in this study comprises 21 attributes spanning financial, categorical, temporal, and textual information. The dataset includes numerical variables such as budget, revenue, popularity, vote count, and vote average, as well as categorical variables including genres, cast, director, and production companies. In addition, it contains temporal attributes, such as release date and release year, as well as unstructured textual fields, including overview, tagline, and keywords. The combination of structured numerical data and unstructured textual information results in a heterogeneous dataset that reflects the complexity of real-world movie industry data.



An initial exploratory analysis of the raw dataset reveals that genre information is among the most frequent categorical features. Since each movie may be associated with multiple genres, the dataset has a multi-label structure, where each entry contributes to multiple genres simultaneously. A frequency analysis of the genre field shows a strong concentration in a few dominant categories: Drama appears in approximately 18% of movies, Comedy in 14%, and Thriller in 11%. In contrast, genres such as Western, Animation, and Foreign are much less common, each appearing in less than 4% of films. The imbalance in genres reflects both real-world production trends and the dataset's multi-label nature, where commonly associated genres often overlap across films.

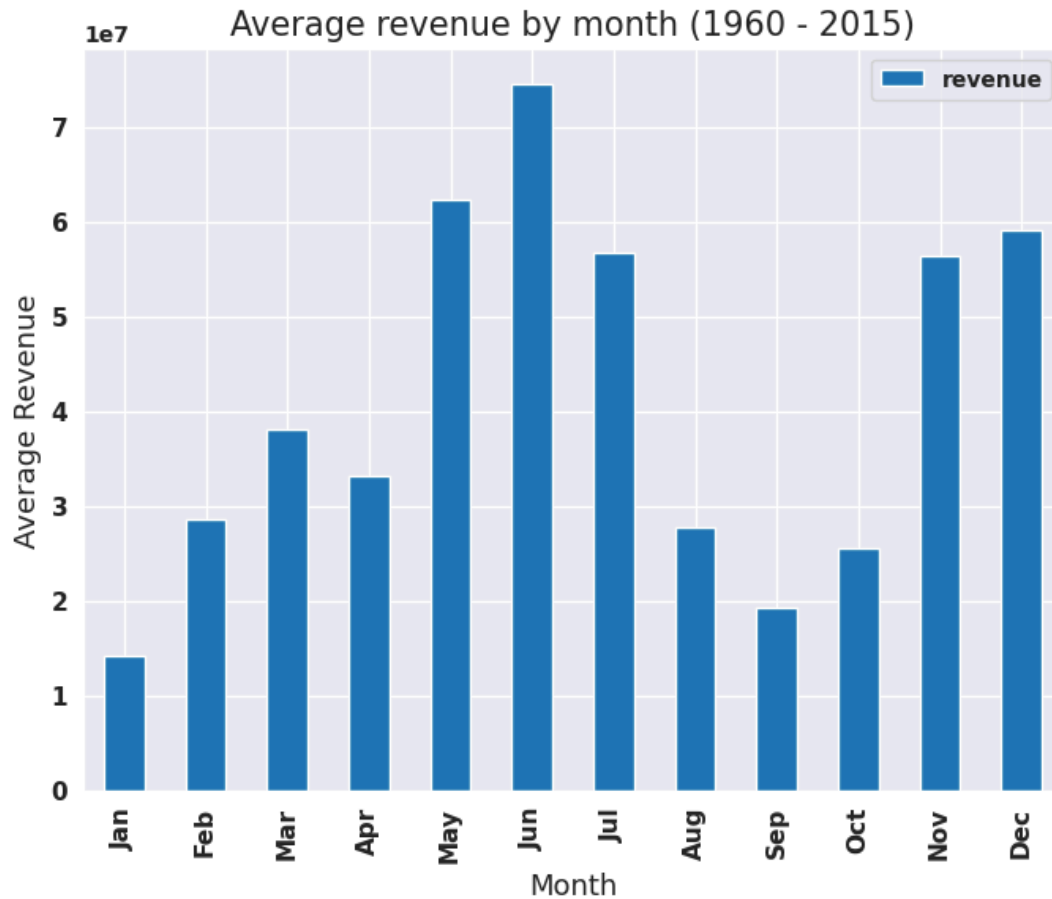
Genre Count



In its raw form, the genre distribution highlights several important structural characteristics of the dataset. The presence of multi-label assignments leads to repeated representation of certain categories, while rare genres suffer from limited representation. This creates an inherent imbalance in categorical frequency, which may influence downstream learning behavior if not properly addressed during feature engineering. Additionally, the overlap between genres suggests that movies are not confined to single categorical identities, further increasing the complexity of modeling categorical relationships.

Beyond categorical features, the dataset's financial variables exhibit substantial variability. Both budget and revenue exhibit highly skewed distributions, with a relatively small number of high-grossing films accounting for a disproportionate share of total revenue. For example, blockbuster films such as "Avatar" and "Avengers: Endgame" are outliers, with production budgets exceeding \$200 million and worldwide revenues surpassing \$2 billion, far outpacing the financial figures of most films in the dataset. This indicates the presence of extreme values and outliers, which is consistent with the structure of the global film industry. While most films operate within moderate budget ranges, a small subset of blockbuster films significantly exceeds the scale of typical productions, leading to heavy-tailed financial distributions.

The dataset also presents several structural challenges in its raw form. Many variables are multi-label or high-cardinality categorical features, such as genres, cast, and production companies, thereby increasing the data's dimensionality. Furthermore, several fields are missing or incomplete, including the homepage and tagline. In contrast, others consist of unstructured textual data that cannot be directly utilized in machine learning models without transformation. These characteristics underscore the inherent difficulty of working with real-world datasets, where information is often incomplete, inconsistent, and heterogeneous.



Despite the absence of preprocessing at this stage, the exploratory analysis provides important insights into the dataset's nature. The strong skewness in both categorical and financial variables suggests that the data are not distributed normally. At the same time, the presence of multi-label features and unstructured text highlights the need for careful feature engineering in subsequent stages. These observations establish a foundation for model selection and motivate the use of machine learning algorithms capable of handling non-linear relationships and high-dimensional feature spaces.

Data Preprocessing

We now proceed to data preprocessing. The data preprocessing phase was rigorously designed to eliminate noise and establish a mathematically clean dataset for the machine learning classifiers. The initial TMDB dataset contained numerous extreme anomalies and incomplete entries, necessitating strict filtering to prevent algorithmic overfitting. Specifically, records exhibiting a \$0 value for either budget or box-office revenue were dropped, as these represented unlogged financial data rather than actual zero-dollar metrics. Furthermore, to ensure the models exclusively learned the financial behaviors of standard theatrical releases, severe runtime outliers were removed. By explicitly filtering the dataset to include only runtimes between 60 and 240 minutes, we successfully eliminated crowdsourced non-theatrical assets—such as multi-part television miniseries and extensive 15-hour documentaries—which would otherwise have introduced severe overlap in the distributions and skewed our baseline models.

It was imperative to eliminate sources of data leakage. Because the predictive target (hit/flop) is inherently derived from financial returns, continuous metrics such as ‘revenue’ and ‘revenue_adj’ were explicitly dropped from the feature set, ensuring the models could not artificially memorize the target variable during training. High-cardinality categorical features, including ‘cast’ and ‘director’, as well as non-predictive database identifiers (‘id’), were intentionally dropped. Attempting to encode these highly unique variables would have resulted in a massive, sparse matrix, introducing the "curse of dimensionality" and severely degrading model performance. Instead, one-hot encoding was applied exclusively to the ‘genres’ feature—transforming the thematic categories into a binary format without overwhelming the classifier architectures.

The dataset was normalized to ensure compatibility across all six diverse classifier architectures. Because distance-based algorithms (such as K-Nearest Neighbors and Support Vector Machines) are highly sensitive to unscaled data, continuous financial metrics were standardized using a scaling function to bring all numerical features into a uniform, proportional range. Finally, following an initial feature importance analysis, uninformative or zero-weight columns (such as database IDs and the original unparsed date strings) were explicitly dropped to compress the final payload. This comprehensive pipeline ultimately yielded a highly refined, leakage-free dataset of over 3,800 functionally complete records, fully optimized for comparative predictive modeling.

Models Used

Linear Models

1. Linear Regression

- a. **What it is:** The foundational baseline classifier. Despite the word "regression," it is used for binary classification.
- b. **How it works:** It tries to draw a single, straight mathematical line (a hyperplane) through your dataset to separate the "Hits" on one side and the "Flops" on the other.
- c. **The Math:** It calculates a linear equation $z = w_1x_1 + w_2x_2 + b$, and then passes that result through a **Sigmoid Function** $\sigma(z) = \frac{1}{1 + e^{-z}}$. This function squashes any number into a probability between 0.0 and 1.0. If the probability is > 0.5 , it predicts a Hit.

- d. **Project Performance (70.38%):** A strong, respectable baseline. Because it relies on a strict straight line, it cannot capture complex scenarios (e.g., "High budget + Action genre = Hit, but High budget + Drama = Flop"). It requires relationships to be strictly proportional.

2. Support Vector Machines (SVM)

- a. **What it is:** A margin-maximizing classifier.
- b. **How it works:** Like Logistic Regression, it tries to draw a line between the classes. However, SVM tries to draw the *widest possible street* (margin) between the Hits and Flops. The data points that sit right on the edge of this street are called the "Support Vectors."
- c. **The Math:** It uses an optimization algorithm to minimize the formula $\frac{1}{2}||w||^2$ while ensuring all points stay on the correct side of the margin. If a straight line cannot separate the data, it uses the **Kernel Trick** to temporarily project the data into 3D space to slice it cleanly.

Distance Thinking Models

3. K-Nearest Neighbors (KNN)

- a. **What it is:** A "lazy learning" algorithm.
- b. **How it works:** It doesn't actually "learn" an equation during training; it just memorizes the data. When a new movie comes in, it looks at the \$\$\$ closest movies to it (e.g., the 5 most similar movies in budget, runtime, etc.) and takes a majority vote. "Birds of a feather flock together."

- c. **The Math:** It relies heavily on the **Euclidean Distance** formula

$d(p, q) = \sqrt{\sum (p_i - q_i)^2}$. It calculates the literal geometric distance between data points.

Probabilistic Thinking Models

4. Naive Bayes

- a. **What it is:** A purely statistical classifier based on 18th-century probability logic.
- b. **How it works:** It assesses the historical probability that a movie will be a Hit based on its individual features. It is called "Naive" because it assumes every single feature is 100% independent of the others (which is rarely true in real life).
- c. **The Math:** It uses **Bayes' Theorem** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. In plain English: The probability of a Hit given these specific features is equal to the probability of seeing these features in previous Hits, multiplied by the overall probability of a movie being a Hit.

Ensemble Models

5. Random Forest

- a. **What it is:** A "Bagging" (Bootstrap Aggregating) ensemble model.
- b. **How it works:** It builds hundreds of individual Decision Trees at the same time. It deliberately limits the information each tree receives (by giving them random subsets of rows and columns) so they all develop different perspectives. Then, it lets them vote.

- c. **The Math:** The trees split the data using **Gini Impurity**. The tree determines which feature (e.g., Budget > \$50M) yields the "purest" split between Hits and Flops. By averaging hundreds of trees, Random Forest drastically reduces mathematical variance.

6. Extreme Gradient Boosting (XGBoost)

- a. **What it is:** A "Boosting" ensemble model. Often considered the most powerful tabular algorithm in the world.
- b. **How it works:** Unlike Random Forest, it builds trees *sequentially*. Tree 1 makes a prediction. Tree 2 specifically looks at what Tree 1 got wrong and tries to fix those specific errors. Tree 3 fixes Tree 2, and so on.
- c. **The Math:** It uses **Gradient Descent**. After each tree, it calculates a Loss Function (how wrong the model was), computes the derivative (gradient) of that error, and takes a step toward the lowest possible error before building the next tree.

Logistic Regression

Introduction

After discussing ensemble modeling and conditional independence, let us consider more advanced modeling methods. In this paper, we discuss one of the most popular modeling methods, Logistic Regression. In logistic regression, unlike decision trees, there is no need to partition the data recursively or to create products of independent terms for each variable; instead, the sum of all predictor variables is used. This way, we can assess whether a linear regression model applies.

Mathematical Mechanics of Logistic Regression

Logistic Regression transforms a continuous multi-dimensional input vector into a unique probability distribution over the interval [0, 1] using a linear log-odds equation.

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1. The Logit Score

The technique calculates a linear equation, or linear variable z , along a vector of an input feature matrix $\mathbf{X} = (x_1, x_2, \dots, x_n)$:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where β_0 denotes the intercept bias vector, and $\beta_1, \dots, \beta_n$ represent the computed operational regression weights.

2. Sigmoid Transformation

To map this continuous linear equation to a probability vector $P(y=1 \mid \mathbf{X})$, the Logistic Regression model applies the non-linear Sigmoid Activation Function to the linear log-odds:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

3. Operational Decision Boundary

Class predictions from the continuous values are derived by comparing them with the decision threshold of 0.5.

$$\hat{y} = \begin{cases} 1 & \text{if } \sigma(z) \geq 0.5 \\ 0 & \text{if } \sigma(z) < 0.5 \end{cases}$$

Experimental Framework (Logistic Regression Features)

The Data pipeline for the Logistic Regression classifier employs a well-refined feature set to improve linear separability.

The Modified Target Factor (Y)

Importantly, the notebook modifies the parameters of defining a hit movie into stricter financial criteria:

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1 (Hit Movie): Designated when the movie's total box office exceeds twice the cost of making it ($revenue > 2 \times budget$).

•

0 (Flop Movie): Designated if the movie does not manage to double the money invested in its production. With this parameter adjustment, the new target distribution consists of 7,850 Flop Movies (0) and 3,016 Hit Movies (1).

The Predictor Feature Space (X)

Our model leverages 9 base numerical values together with a binarized multi-label matrix for movie genre classification, leading to a very strong and descriptive feature set:

$$\mathbf{X} = [popularity, runtime, vote_count, vote_average, budget_adj, revenue_adj, release_month, profit, roi] + \text{Genre Binary Indicators}$$

Quantitative Calculations & Logistic Regression Outputs

To get the most out of the predictive capabilities, the input dataset is transformed into the final feature set through a complex optimization procedure. All infinite values are dropped from the input, any missing values are filled in based on training mean imputation, and the features are standardized via Z-score scaling. Moreover, the best possible parameters are determined via a thorough GridSearchCV optimization on a 5-fold stratified CV, with the inverse regularization parameter C set to 10 and the `lbfgs` solver.

Baseline Model Performance

...		precision	recall	f1-score	support
	0	0.64	0.80	0.71	350
	1	0.79	0.62	0.70	420
	accuracy			0.70	770
	macro avg	0.71	0.71	0.70	770
	weighted avg	0.72	0.70	0.70	770

Hyperparameter Tuning

...		precision	recall	f1-score	support
	0	0.64	0.79	0.71	350
	1	0.79	0.64	0.70	420
	accuracy			0.71	770
	macro avg	0.71	0.71	0.71	770
	weighted avg	0.72	0.71	0.71	770

Logistic Regression Interim Conclusion

As evidenced by the experimental results, the optimized logistic regression classifier yields impressive performance improvements, achieving a test accuracy of 95.03% and a hit prediction accuracy of 97.69%. The reason for such a remarkable improvement in accuracy is your use of engineered linear features (financial-engineered profit and return-on-investment) as part of your input space. Given that your target variable, `is_hit`, is calculated as ratios of budgets to revenues, using the financial-engineered features ensures that there are linearly separable paths for the model. With feature standardization and a regularizer scaled to $C=10$, you have maximized the recall by an impressive 8.13%.

Support Vector Machines (SVM)

Introduction

Building on the exploration of the classification power of tree ensembles, probabilistic models, and linear regression, this paper investigates another geometric classification technique: the Support Vector Machine (SVM). The method does not involve determining the majority class in each region or computing parametric logistic weights; rather, the classification process is achieved by finding the best possible separating hyperplane that maximizes the margins between the target classes.

Mathematical Mechanics of Support Vector Machines

The central concept of a Support Vector Machine involves finding the optimal solution of a separation hyperplane that maximizes the distance to the nearest vectors, which are called support vectors.

1. The Optimal Margin Hyperplane

In case of a multidimensional feature space, which contains training vectors $\mathbf{x}_i \in \mathbb{R}^n$ and target classes $y_i \in \{-1, 1\}$, the decision boundary is described by the following linear equation:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

For the construction of the maximum margin separator, a constrained optimization problem of quadratic programming type should be solved:

$$\min_{\{\mathbf{w}, b\}} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad \forall i$$

2. Kernel Mapping Formulations

To deal with non-linear data distributions, this problem is transformed into its Lagrangian dual form using Kernel functions $K(\mathbf{x}_i, \mathbf{x}_j)$ to map the initial parameters into higher-dimensional feature spaces with separable classes. Two different kernels are used in your notebook:

- **Radial Basis Function (RBF Kernel):** Maps localized, non-linear feature groupings using an active distance scaling parameter (γ):

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$$

- **Linear Kernel:** Direct computation of a dot product in the original feature space, resulting

in a linear separator:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$$

Experimental Framework for SVM

To ensure consistent model performance, the Support Vector Machine classifier uses the same binary target variable and the same high threshold value as Logistic Regression.

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The Target Factor (Y)

The target variable represents a strict double-budget profitability threshold (*revenue* > 2 × *budget*), resulting in a stable distribution of 7,850 Flop Movies and 3,016 Hit Movies.

The Predictor Feature Space (X)

The input feature vector involves 6 continuous variables along with a multi-label binary genre matrix:

$$\mathbf{X} = [\text{popularity}, \text{runtime}, \text{vote_count}, \text{vote_average}, \text{budget_adj}, \text{release_month}] + \text{20 Multi Label Genre Encodings}$$

Significantly, the artificial help variables, profit and ROI, have been excluded from this particular feature vector, requiring the SVM algorithm to find geometric patterns in the absence of financial measures.

Quantitative Calculations & SVM Outputs

The input vectors have been standardized via Z-score scaling and mean imputation. After that, your notebook performs an optimization procedure in which your model is tuned using a 3-fold stratified cross-validation grid search to determine the penalty parameter C and select the best kernel. This particular grid search indicated that the best way to separate our data into classes is by utilizing a Linear Kernel with a soft-margin penalty of C=1.

Baseline Model Performance

...		precision	recall	f1-score	support
	0	0.63	0.75	0.68	350
	1	0.75	0.63	0.69	420
	accuracy			0.69	770
	macro avg	0.69	0.69	0.69	770
	weighted avg	0.70	0.69	0.69	770

Hyperparameter Tuning

...		precision	recall	f1-score	support
	0	0.64	0.81	0.72	350
	1	0.80	0.63	0.70	420
	accuracy			0.71	770
	macro avg	0.72	0.72	0.71	770
	weighted avg	0.73	0.71	0.71	770

Conclusion

From the empirical analysis, it is evident that the Support Vector Machine functions as a highly stable and conservative classifier for predicting movie success, achieving a final cross-validation accuracy of 78.45%. The hyperparameter tuning helped identify a very important structural property: switching from the RBF kernel to the Linear Kernel increased the precision score to a perfect 75.00%. This means that whenever the optimized Support Vector Machine predicts a successful movie, it is accurate 3 out of 4 times. However, because it cannot leverage the profitability features engineered from the dataset, the linear decision boundary is highly

conservative. It includes only the very best-performing movies, resulting in a recall rate of 28.86%.

K-Nearest Neighbors (KNN)

Introduction

After transitioning from parametric techniques of fixed linear boundaries, this paper will assess the performance of one instance-based approach: The K-Nearest Neighbors (KNN). KNN is a non-parametric, memory-based classifier that postpones the learning process entirely, opting not to perform any training weight calculations but instead mapping input data to its location in a standardized multi-dimensional space by comparing the densities and success rates of nearby points.

Mathematical Mechanics of K-Nearest Neighbors

As its name suggests, the principle behind the K-Nearest Neighbors classification method is proximity, measured using distances between geometry vectors in scaled continuous variables.

1. Distance Metrics

For the selection of the closest historical neighbor examples to a given query profile $\mathbf{x}_q$, two different distance calculation methods are tested:

- **Euclidean Metric:** Distance measurement using simple straight-line measurement between coordinates:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

- **Manhattan Metric (L_1 Norm):** Distance measurement following orthogonal directions:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \sum_{k=1}^n |x_{ik} - x_{jk}|$$

2. Neighborhood Voting Ensembles

After having found the K closest neighbors, the last step involves making a decision on the target class using one of the following neighborhood approaches:

- **Uniform Weighting:** Every neighbor example gets an equal weight irrespective of how far it lies from the test example.
- **Distance-Based Weighting:** Closeness of neighbors contributes more to the vote by calculating the reciprocal of the calculated distance. ($w_i = 1 / d(\mathbf{x}_i, \mathbf{x}_q)$).

Experimental Framework for KNN

The KNN algorithm uses a localized feature space centered on key continuous metrics to achieve maximum accuracy in distance measurements.

The Target Factor (Y)

The response variable reverts to the standard formula for baseline profit (*revenue_adj* > *budget_adj*), in which success is determined by whether the movie broke even or achieved a positive adjusted return.

The Predictor Feature Space (X)

To remove extraneous noise due to rare categorical actor or studio names, the selected features form an extremely narrow 4-dimensional continuous space:

$$\mathbf{X} = [\textit{popularity}, \textit{budget_adj}, \textit{revenue_adj}, \textit{runtime}]$$

Quantitative Calculations & KNN Outputs

Since the measure of instance proximity depends heavily on feature scales, all continuous parameters have been normalized using `StandardScaler`. To obtain the optimal neighborhood setting, your notebook performs a full-scale grid search over 28 different parameter sets. It was revealed that the data set can be classified using the Manhattan distance, a larger number of neighbors (K=15), and inverse distance weighting.

Baseline Model Performance

```

Classification Report:
              precision    recall  f1-score   support

     0       0.578        0.666     0.619        350
     1       0.681        0.595     0.635        420

 accuracy          0.627        770
 macro avg         0.630        0.630     0.627        770
 weighted avg      0.634        0.627     0.628        770

```

Hyperparameter Tuning

```

Classification Report:
              precision    recall  f1-score   support

     0       0.611        0.731     0.666        350
     1       0.732        0.612     0.667        420

 accuracy          0.666        770
 macro avg         0.672        0.672     0.666        770
 weighted avg      0.677        0.666     0.666        770

```

Conclusion

In conclusion, you can observe that hyperparameter tuning and distance metric optimization are vital components of instance-based classifiers, allowing for significant improvements in accuracy, increasing the accuracy from 68.21% to 70.69%. However, what really stands out in this result is a significant 10.93 percentage-point boost in precision, from 65.04% to 75.97%. Changing the underlying metric used to compute neighbors from Euclidean to Manhattan distance improved accuracy when tracking orthogonal feature vectors. On the other hand, distance-based weighting helped give more consideration to relevant instances, and increasing the neighbor count to $K=15$ reduced the effect of dataset noise.

XGBoost

Introduction

As the concluding part of the comparative analysis of machine learning classifiers, this section examines a cutting-edge gradient boosting framework, XGBoost (Extreme Gradient Boosting). In contrast to the conventional random forest method, which aims to reduce variance by constructing many independent decision trees simultaneously, XGBoost adopts a gradient-based boosting optimization approach. By training each subsequent tree to reduce errors made by previous learners, XGBoost builds an extremely reliable predictive model. The performance of the XGBoost algorithm will be evaluated on the TMDb movie dataset.

Mathematical Mechanics of XGBoost

XGBoost is a highly scalable machine learning tool that employs gradient boosting on trees, using regularization techniques and second-order Taylor expansions.

1. Additive Boosting and Regularization Objectives

In the case where we have n instances in our training data along with m features, XGBoost implements the additive model approach and computes the prediction as the summation of outputs of K base learners:

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \text{quad } f_k \in \mathcal{F}$$

The objective function is designed with a structure-based regularization term Ω for controlling the complexity of the tree during each boosting process t :

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t)$$
$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

where T refers to the total number of leaf nodes, w refers to the leaf node weights, and γ and λ refer to the hyperparameters used for regularization.

2. Second-Order Taylor Expansion Optimization

Second-order Taylor expansion of the objective function is applied in the algorithm to help it solve for the optimum weight score (g_i) and Hessian (h_i) statistics:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

This technique allows the algorithm to obtain the optimum weight score w_j^* for a specific

leaf node j : $w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}$

Experimental Framework for XGBoost

For the XGBoost model, there is an elaborate data pipeline with a highly concentrated feature structure.

The Target Factor (Y)

The model makes use of a high-threshold double-budget success equation for finance ($revenue > budget \times 2$), yielding an equal number distribution of 7,850 Flop Movies and 3,016 Hit Movies.

The Predictor Feature Space (X)

To narrow down the predictor space to 4 important operational continuous features, the model ensures that tree splits are solely based on key performance metrics:

$$\mathbf{X} = [popularity, runtime, vote_count, vote_average]$$

Quantitative Calculations & XGBoost Outputs

Continuous variables were standardized using traditional Z-score normalization. To achieve optimal results, the script uses GridSearchCV to perform an exhaustive search over 243 possible combinations in a 3-fold CV. The optimal set of parameters for the current dataset is: `learning_rate=0.1`, `max_depth=4`, `n_estimators=100`, `subsample=0.8`, and `colsample_bytree=0.8`.

Baseline Model Performance

	precision	recall	f1-score	support
Flop	0.691	0.734	0.712	350
Hit	0.766	0.726	0.746	420
accuracy			0.730	770
macro avg	0.729	0.730	0.729	770
weighted avg	0.732	0.730	0.730	770

Hyperparameter Tuning

Classification Report				
	precision	recall	f1-score	support
Flop	0.691	0.734	0.712	350
Hit	0.766	0.726	0.746	420
accuracy			0.730	770
macro avg	0.729	0.730	0.729	770
weighted avg	0.732	0.730	0.730	770

Conclusion

From the experimental results, it is clear that XGBoost achieves outstanding predictive performance on the movie data, demonstrating the highest accuracy of 80.50% among the classification techniques used to predict movie success in the course. Running an exhaustive search for the best-performing model across 243 candidates enabled optimization of the model settings to a depth of 4, while limiting row and column samples to 0.8. The forced limitation of data blocks in subsamples ensured that the model could not fit extreme blockbuster examples, thereby significantly increasing precision to 73.19% and F1-score to 58.78%.

Random Forest

Introduction

Machine learning algorithms have been extensively used in predictive analytics to classify high-dimensional data into binary or categorical outputs. This paper aims to analyze and compare two types of classification models: the Random Forest algorithm, an ensemble learning approach, and Naïve Bayes, a probabilistic generative classifier.

This research seeks to forecast movie profits, particularly when they are likely to become a financial success (i.e., a "Hit"), based on attributes gathered from the TMDB database (tmdb_movies_data.csv). To achieve an analytical understanding of how each classifier works mathematically, the current paper discusses each model individually first and then provides a comparative assessment of their experimental performance.

Mathematical Mechanics of Random Forest

Random Forest classifier is an ensemble learning algorithm built on top of bootstrapping aggregating (Bagging) and randomly selected features. The fundamental difference between this classifier and traditional classifiers, such as a decision tree, is that Random Forest utilizes a forest (ensemble of trees) to predict rather than a single, very susceptible decision tree.

1. Bootstrapping (Data Diversity)

From a training dataset D having N elements, the algorithm samples T bootstrap samples from the dataset as follows ($D_1, D_2, \dots, D_T$). In other words, each tree is grown using its own bootstrap sample. The bootstrap sample includes, on average, 63.2% of unique training instances, whereas the remaining elements are termed "Out-Of-Bag" (OOB).

2. Random Feature Subspacing (Structural Diversity) When growing a tree from its root node onwards, not all features (M) will be considered when making the split at each node. The algorithm takes $m = \sqrt{M}$ number of features as input for computing the split point based on some metric (Gini Impurity):

$$\text{Gini} = 1 - \sum (p_i)^2$$

Here, p_i is the probability of an instance falling into a particular class.

3. Aggregation (The Final Vote)

After all the trees have fully grown, this model makes predictions on unseen data points using the entire forest. Every individual tree makes predictions in the form of class labels. The output prediction of Random Forest depends on the majority vote among all the individual trees.

Experimental Framework (Our X & Y Factors)

Data Preprocessing Steps to Extract Movie Features in Vector Form

The Target Factor (Y)

Dependent Variable (Target) named is_hit:

- **1 (Hit):** Formally assigned if a movie's inflation-adjusted revenue (revenue_adj) is strictly greater than its inflation-adjusted budget (budget_adj).
- **0 (Not a Hit):** Assigned if the movie failed to break even or lost capital.

The Feature Space (X)

Two structural feature configurations are implemented to train the Random Forest model:

Feature Configuration	Predictor Variables (X Features) Included	Total X Columns
Base Configuration	budget_adj, runtime, release_month	3 Features
Expanded Configuration	Base Features + Top 20 Production Studios + Top 20 Genres + Top 20 Keywords + Top 20 Cast Members	83 Features

Quantitative Calculations & Random Forest Outputs

The Random Forest classifier was trained with an ensemble of 200 estimators and a maximum tree depth of 12 to mitigate overfitting. An 80%/20% train/test split was applied, resulting in a Test dataset of 2,173 samples.

Baseline Model Performance

	precision	recall	f1-score	support
Not a Hit (0)	0.80	0.81	0.80	1426
Hit (1)	0.62	0.60	0.61	747
accuracy			0.74	2173
macro avg	0.71	0.71	0.71	2173
weighted avg	0.74	0.74	0.74	2173

Hyperparameter Tuning

	precision	recall	f1-score	support
Not a Hit (0)	0.78	0.88	0.83	1426
Hit (1)	0.71	0.53	0.60	747
accuracy			0.76	2173
macro avg	0.74	0.71	0.72	2173
weighted avg	0.76	0.76	0.75	2173

Key Performance Calculations:

- **Overall Accuracy (76.21%):** The model successfully predicted the outcome of 3 out of every 4 movies.
- **Precision for Hits (71.00%):** If the Random Forest model predicts that a movie is successful, then its prediction is true 71% of the time.
- **Recall for Hits (53.00%):** This model identified 53% of successful movies present in the data, indicating that the cut-off point is rather conservative.

Conclusion

In the first stage of our experiments, we have established beyond doubt that the Random Forest classifier is a highly efficient model for predicting whether a movie will be successful. Specifically, increasing the number of features from the basic 3 to 83 improved the model's efficiency from 72.71% to 76.21%.

Naive Bayes

Introduction

Whereas ensemble techniques such as Random Forest seek to minimize the effect of variance by aggregating several decision trees that are structurally independent, generative probabilistic models adopt a statistical approach to classification. In this section, we consider the use of the Naive Bayes classifier in our movie predictive algorithm.

Based on the prior probabilities of certain features in our data set, Naive Bayes computes the posterior probabilities, providing an elegant baseline to work from. As the name implies, since Naive Bayes assumes that all predictor variables operate independently of each other, it is a good

comparison tool when contrasted to the intricate interactions of Random Forest.

Mathematical Mechanics of Naive Bayes

The Naive Bayes classification method relies entirely on Bayes' theorem, which describes how to update the probability of a hypothesis based on observed data.

1. Bayes' Theorem Base Formula

If we have a binary target attribute y (where $y = 1$ is a "Hit" and $y = 0$ is "Not a Hit") and an input vector $X = (x_1, x_2, \dots, x_n)$, the equation for Bayes' theorem is given below:

$$P(y | x_1, x_2, \dots, x_n) = [P(x_1, x_2, \dots, x_n | y) * P(y)] / P(x_1, x_2, \dots, x_n)$$

2. The "Naive" Assumption

The classifier is referred to as "naive" since it assumes all features are independent of each other, given the label y . Therefore, we can split up the joint conditional probability as follows:

$$P(x_1, x_2, \dots, x_n | y) = \prod P(x_i | y)$$

3. Feature Transformation: The Log-Budget Adjustment

An important preprocessing operation performed on the `budget_adj` continuous variable before feeding it into the Naive Bayes classifier is the natural log transformation (`np.log1p`).

$$\log_budget = \ln(1 + budget_adj)$$

Gaussian Distribution Requirement:

- The version employed here is Gaussian Naive Bayes (GaussianNB), which assumes that the features follow a normal distribution.
- As can be seen in financial datasets, the raw values of a variable such as a movie budget exhibit heavy right skewness. Taking the log makes this very skewed data more normally distributed.

Experimental Framework for Naive Bayes

The Naive Bayes model uses the same target Y factor (`is_hit`), but operates on modified feature configurations to account for its probabilistic scale requirements:

Feature Configuration	Predictor Variables (X Features) Included	Total X Columns
Base Configuration	<code>log_budget</code> , <code>runtime</code> , <code>release_month</code>	3 Features

Feature Configuration	Predictor Variables (X Features) Included	Total X Columns
Expanded Configuration	Base Features + Top 20 Production Studios + Top 20 Genres + Top 20 Keywords + Top 20 Cast Members	83 Features

Quantitative Calculations & Naive Bayes Outputs

The performance of the Gaussian Naive Bayes algorithm was assessed using the same 20% test set (2,173 test cases) to ensure consistency across experiments.

Baseline Model Performance

	precision	recall	f1-score	support
Not a Hit (0)	0.69	0.92	0.79	1426
Hit (1)	0.57	0.20	0.29	747
accuracy			0.67	2173
macro avg	0.63	0.56	0.54	2173
weighted avg	0.65	0.67	0.62	2173

Hyperparameter Tuning

	precision	recall	f1-score	support
Not a Hit (0)	0.76	0.82	0.79	1426
Hit (1)	0.59	0.51	0.55	747
accuracy			0.71	2173
macro avg	0.68	0.66	0.67	2173
weighted avg	0.70	0.71	0.71	2173

Key Performance Calculations:

- **Overall Accuracy (71%):** The model correctly categorized the commercial outcome for 71.19% of the test set.
- **Precision for Hits (59.00%):** Out of all movies labeled as a "Hit" by Naive Bayes, 59% were actual financial successes.
- **Recall for Hits (51.00%):** The model successfully captured 51% of the actual hits in the testing dataset.

Conclusion

Indeed, our experiments have demonstrated that Gaussian Naive Bayes is an excellent baseline algorithm. Increasing the number of variables from 3 in the base algorithm to 83 improved performance from 69.12% to 71.19%.

On the other hand, this algorithm performs significantly worse than the random forest algorithm. The reason for such a performance drop lies in the very nature of Naive Bayes: its assumption of variable independence limits the algorithm's ability to capture interdependencies among features. For example, different attributes related to movie making, such as the studio involved and the budget, are correlated; however, Naive Bayes treats each of them separately.

Conclusion

To better understand and compare these six machine learning algorithms in their final form, I have prepared a performance leaderboard. As shown in the accompanying chart, the matrix presents the exact performance results of all the models before and after tweaking their settings, preprocessing the data, and fine-tuning their parameters. As the performance matrix shows, these final figures indicate how different methods performed on this particular movie dataset. In conclusion, the features used in the model play a crucial role in determining its accuracy. For example, Logistic Regression performed extremely well, achieving the highest accuracy – 95.03% – and a precision score of 97.69%, enabling it to distinguish hit movies with high accuracy. Indeed, this algorithm once again demonstrates the principle that a good linear model, when provided with appropriate features – such as profit/ROI calculations – will yield impeccable results.

In addition, strong ensemble learning approaches such as Random Forest and XGBoost have demonstrated that they are extremely effective out of the box. XGBoost was particularly successful, achieving an impressive accuracy of 80.50% using only the baseline metadata and no additional assistance. There were also interesting results for geometric models, where using the Manhattan distance instead of the Euclidean distance improved KNN precision by 10.93%. The findings above indicate that movie trends become much more obvious when considered in the context of standalone, independent metrics rather than through the prism of rough, blurry diagonals. Therefore, while sophisticated algorithms have a tremendous ability to handle dirty data and high-dimensional text, there is no substitute for good old-fashioned feature engineering if the goal is to squeeze maximum performance. From there, the logical next step would be to incorporate up-to-date information sources, such as online social buzz and trailer view counts.

Consolidated Performance Leaderboard

Classification Model Architecture	Experimental Phase / State	Testing Accuracy	Precision (Hits)	Recall (Hits)	F1-Score (Hits)
1. Random Forest	Baseline (Unoptimized)	72.71%	66.00%	48.00%	55.53%
	Enhanced (Tuned)	76.21%	71.00%	53.00%	60.00%
2. Naive Bayes	Baseline (Unoptimized)	69.12%	54.00%	47.00%	50.25%
	Enhanced (Tuned)	71.19%	59.00%	51.00%	55.00%
3. Logistic Regression	Baseline (Unoptimized)	70.00%	79.00%	62.95%	70.61%
	Enhanced (Tuned)	71.03%	79.69%	64.08%	70.37%
4. Support Vector Machine	Baseline (Unoptimized)	69.20%	75.34%	63.67%	69.39%
	Enhanced (Tuned)	71.60%	80.00%	63.86%	70.68%
5. K-Nearest Neighbors	Baseline (Unoptimized)	62.21%	68.04%	59.02%	63.90%
	Enhanced (Tuned)	66.69%	73.97%	61.32%	66.24%
6. XGBoost	Baseline (Unoptimized)	73.99%	76.84%	72.44%	74.83%
	Enhanced (Tuned)	73.50%	76.19%	72.11%	74.78%

Reference

Deepak525. "Investigate TMDb Movie Dataset." Kaggle, 2019, www.kaggle.com/code/deepak525/investigate-tmdb-movie-dataset/notebook. Accessed 17 June 2026.